



US012412827B2

(12) **United States Patent**
Lee et al.

(10) **Patent No.:** **US 12,412,827 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **SEMICONDUCTOR DIE PACKAGE WITH CONDUCTIVE LINE CRACK PREVENTION DESIGN**

(71) Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.**,
Hsinchu (TW)

(72) Inventors: **Ya-Huei Lee**, Zhunan Township, Miaoli County (TW); **Shu-Shen Yeh**, Taoyuan (TW); **Kuo-Ching Hsu**, Taipei (TW); **Shyue-Ter Leu**, Hsinchu (TW); **Po-Yao Lin**, Zhudong Township, Hsinchu County (TW); **Shin-Puu Jeng**, Hsinchu (TW)

(73) Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/513,866**

(22) Filed: **Nov. 20, 2023**

(65) **Prior Publication Data**

US 2024/0096778 A1 Mar. 21, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/377,620, filed on Jul. 16, 2021, now Pat. No. 11,854,956.

(51) **Int. Cl.**
H01L 23/498 (2006.01)
H01L 23/00 (2006.01)

(52) **U.S. Cl.**
CPC .. **H01L 23/49838** (2013.01); **H01L 23/49822** (2013.01); **H01L 24/16** (2013.01);
(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,309,435 B2 * 11/2012 Kaltalioglu H01L 23/585
257/620
8,354,734 B2 * 1/2013 Wang H01L 23/562
257/E21.523

(Continued)

FOREIGN PATENT DOCUMENTS

TW 200428604 A 12/2004

OTHER PUBLICATIONS

Chinese language office action dated Mar. 25, 2023, issued in application No. TW 111110331.

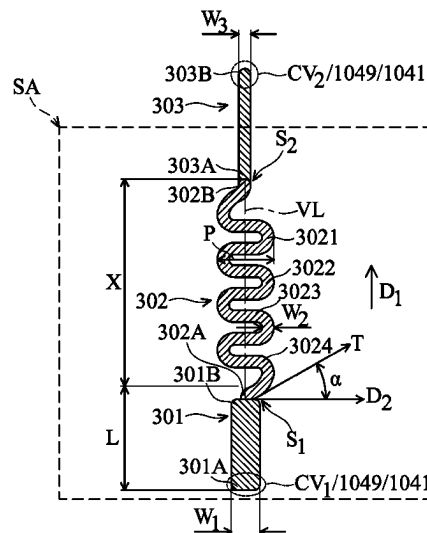
Primary Examiner — Long Pham

(74) *Attorney, Agent, or Firm* — McClure, Qualey & Rodack, LLP

(57) **ABSTRACT**

A semiconductor die package is provided. The semiconductor die package includes a semiconductor die and a package substrate supporting and electrically connected to the semiconductor die. The semiconductor die has a corner. The package substrate includes several conductive lines, and one of the conductive lines under the corner of the semiconductor die includes a first line segment and a second line segment connected to the first line segment. The first line segment is linear and extends in a first direction. The second line segment is non-linear and has a varying extension direction.

20 Claims, 9 Drawing Sheets



(52) **U.S. Cl.**

CPC *H01L 2224/16227* (2013.01); *H01L*
2924/1016 (2013.01); *H01L 2924/20645*
(2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,714,426	B2 *	7/2020	Hsieh	H01L 21/4857
2011/0244658	A1	10/2011	Kaltalioglu et al.	
2022/0361338	A1 *	11/2022	Hsu	H01L 25/0655

* cited by examiner

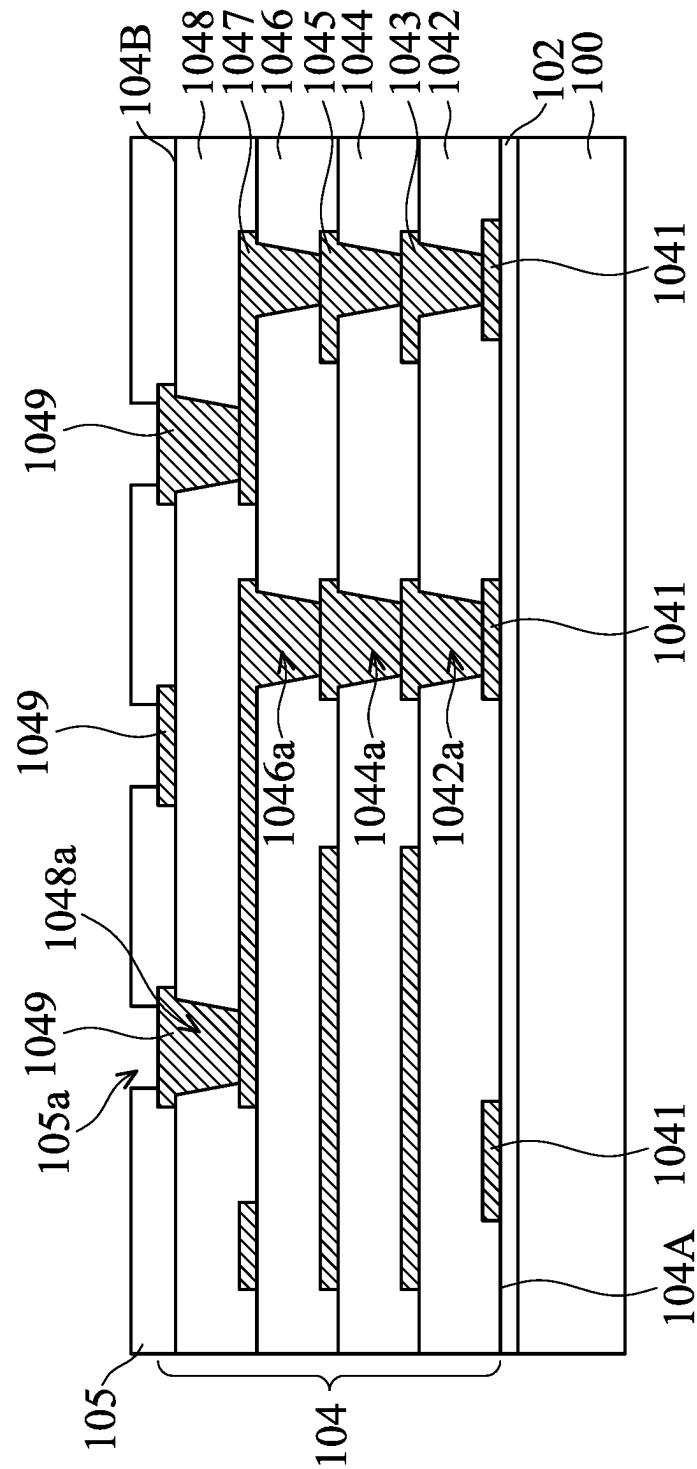


FIG. 1A

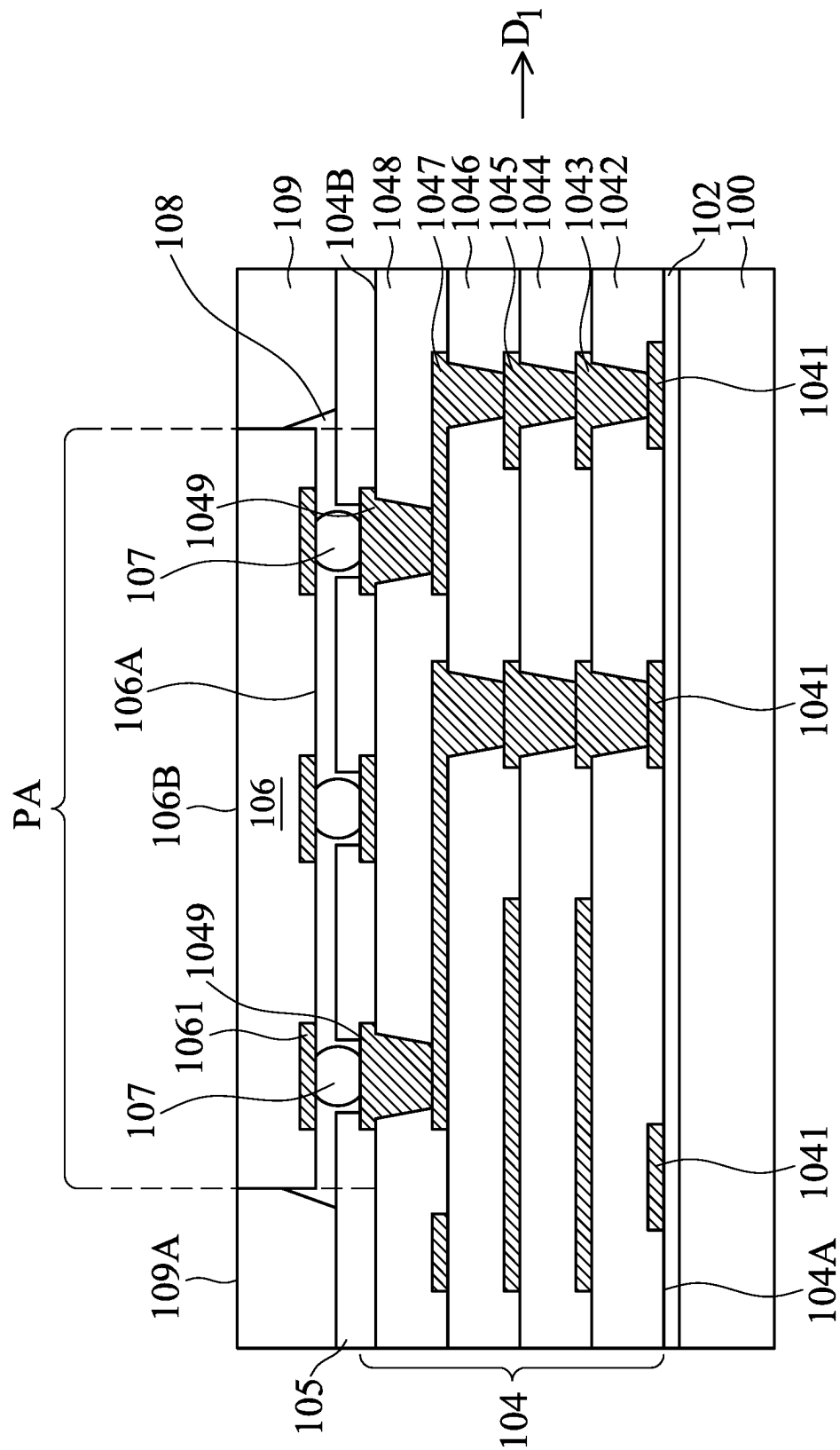


FIG. 1B

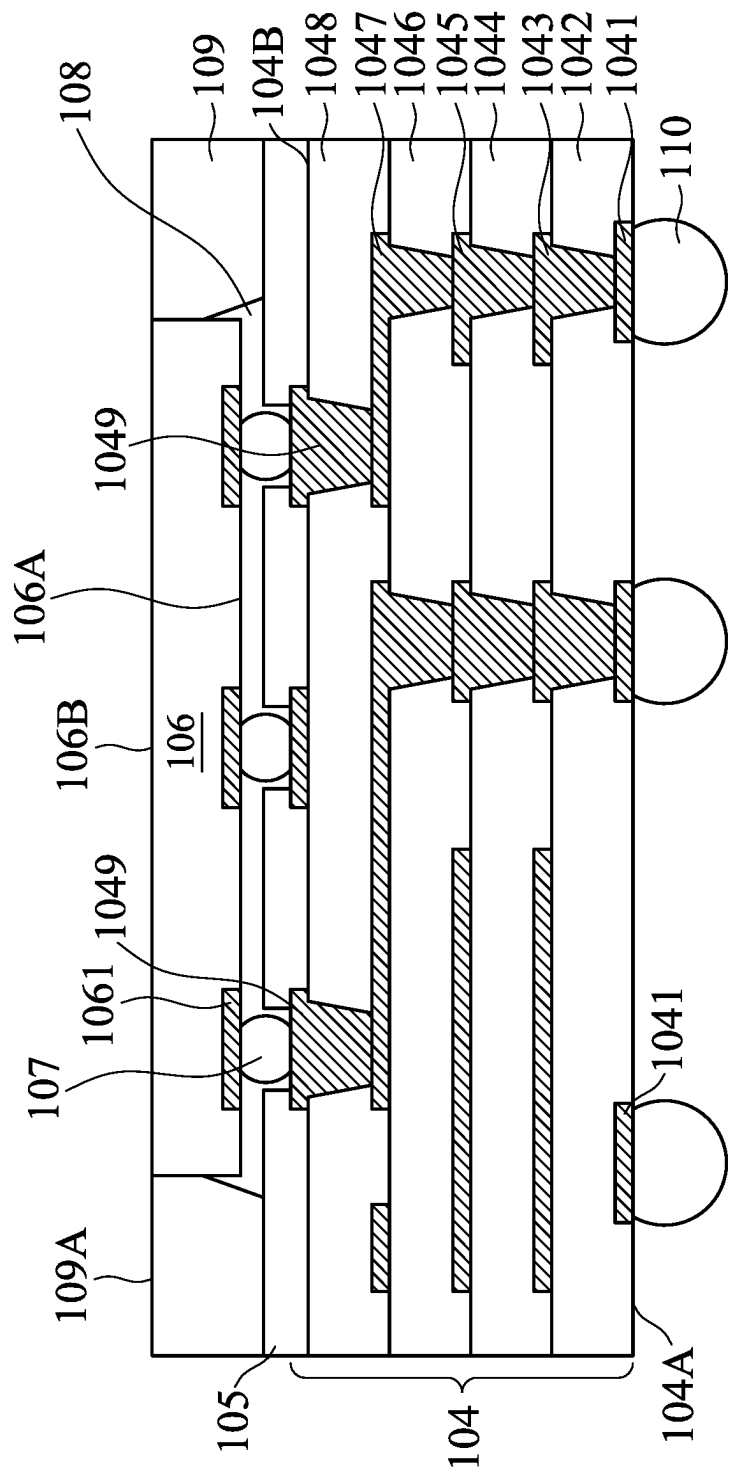


FIG. 1C

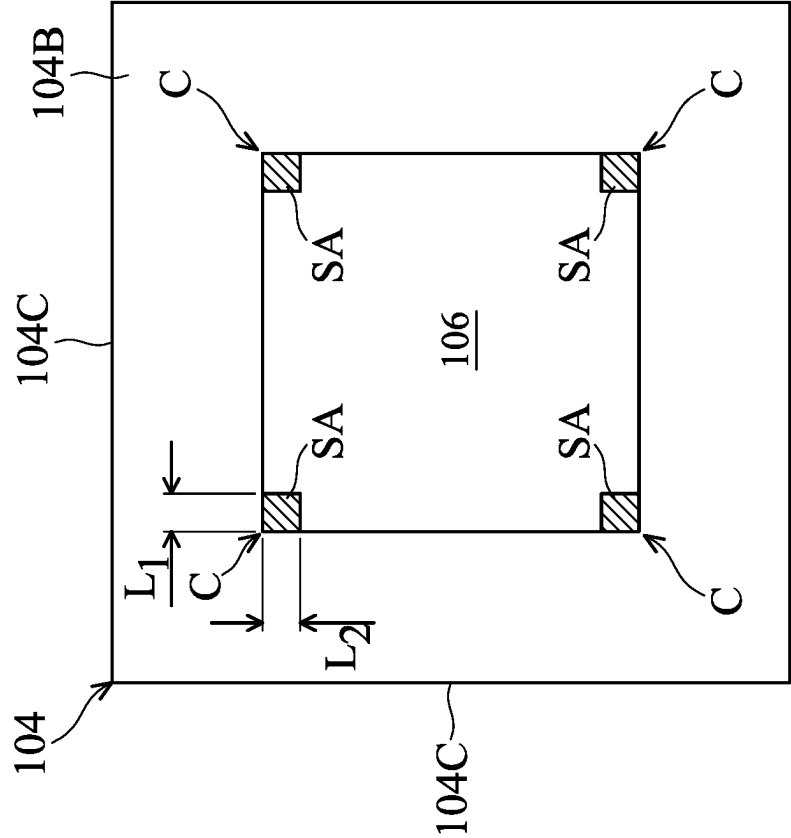


FIG. 2

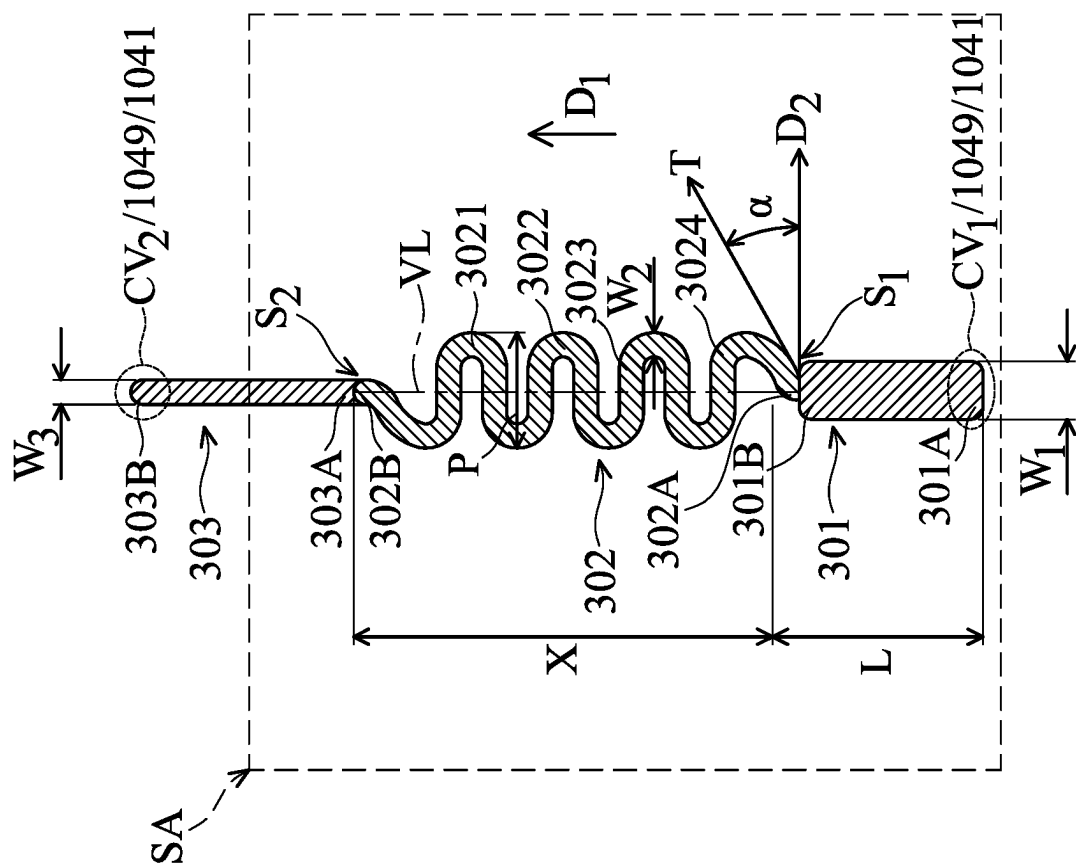


FIG. 3

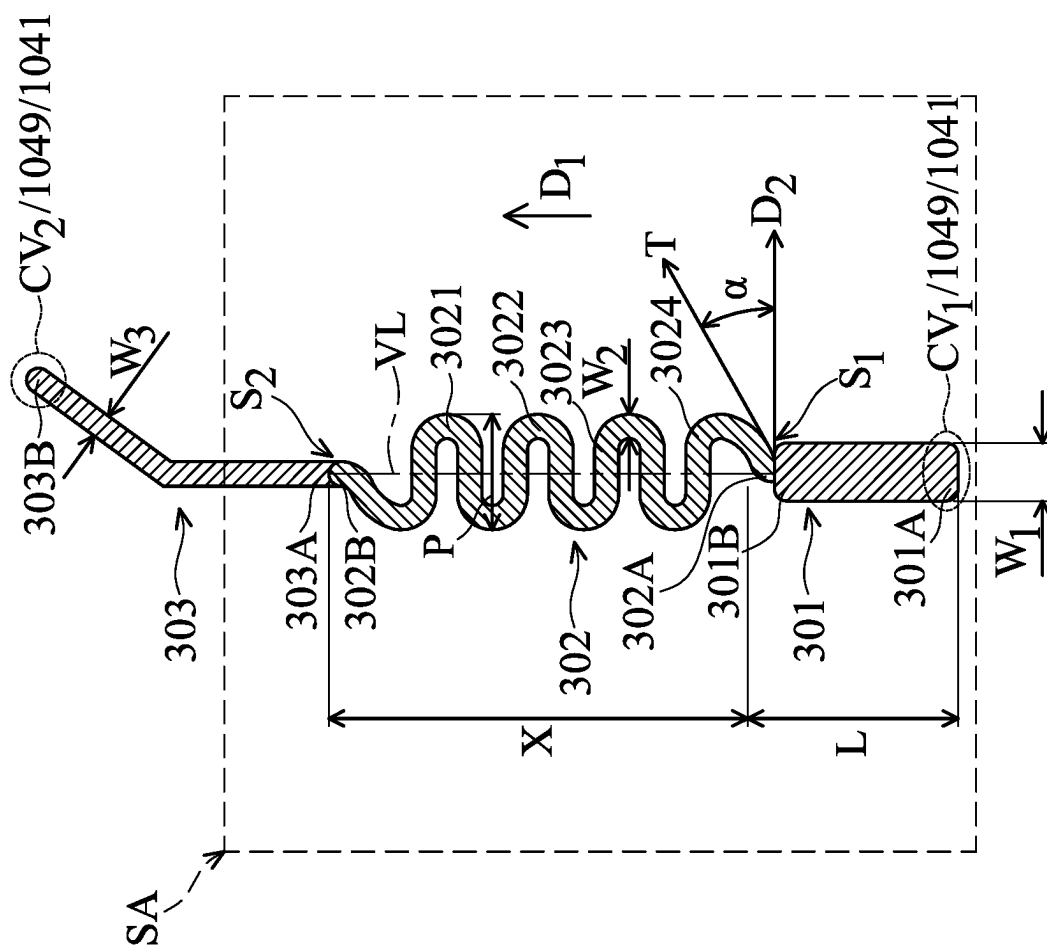


FIG. 3A

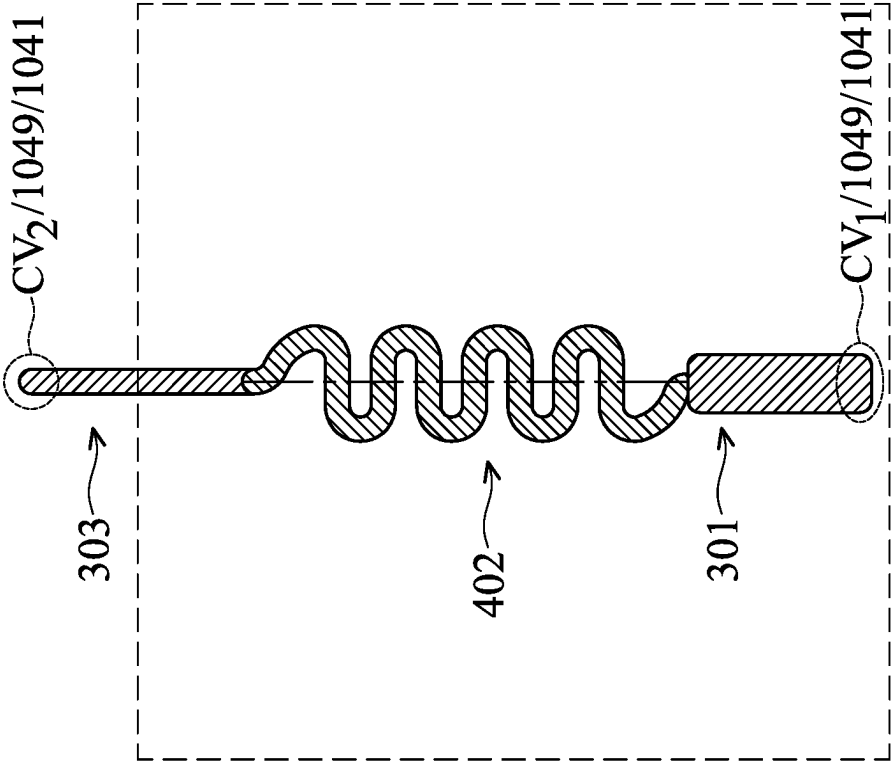


FIG. 4

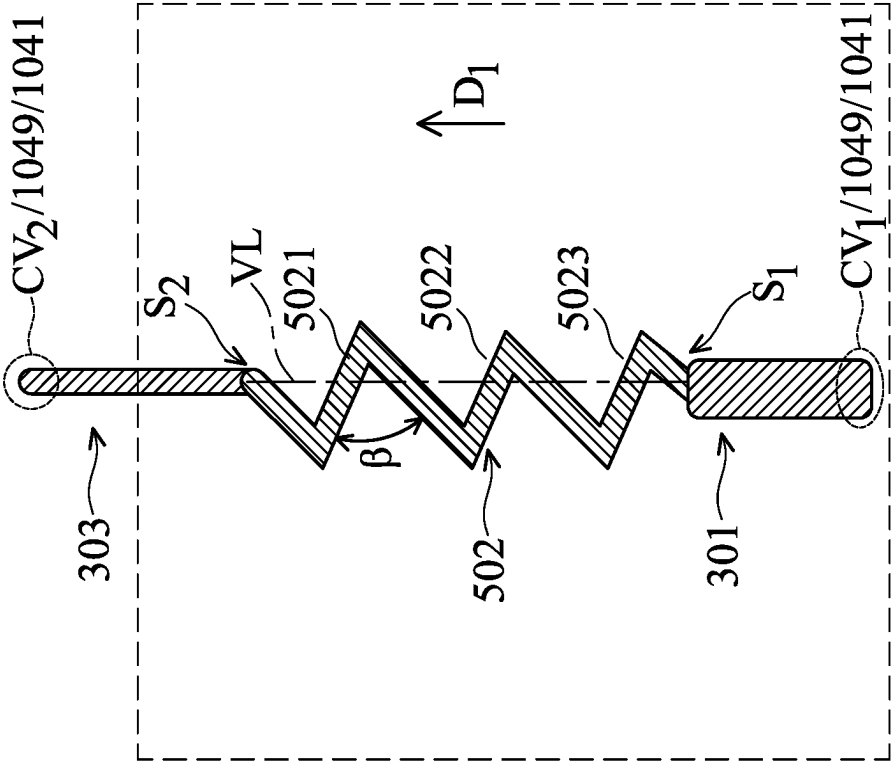


FIG. 5

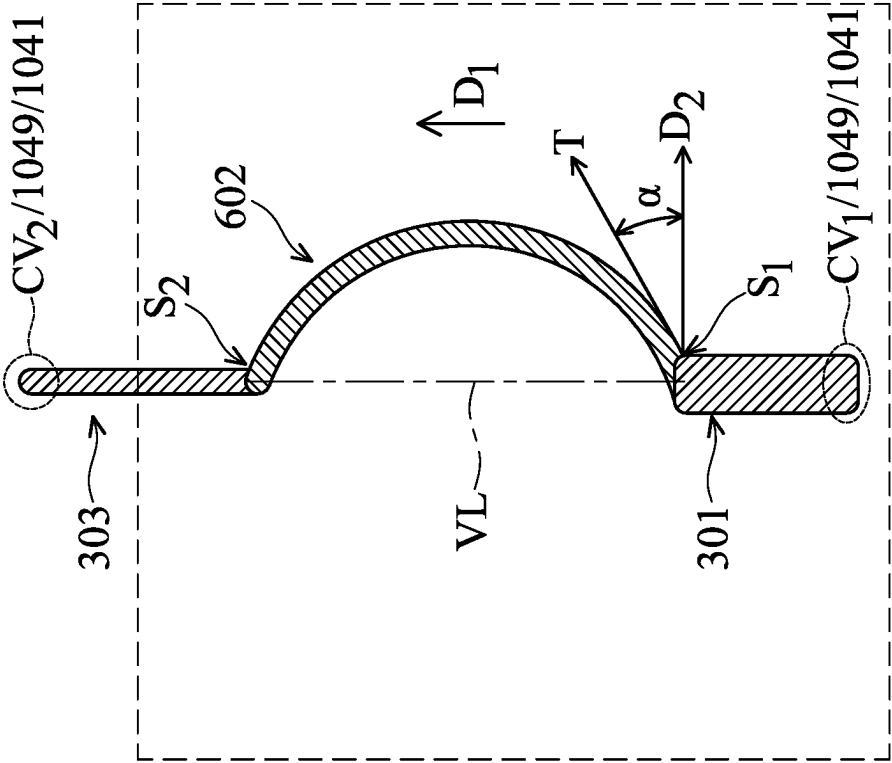


FIG. 6

1

SEMICONDUCTOR DIE PACKAGE WITH CONDUCTIVE LINE CRACK PREVENTION DESIGN

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a Continuation of U.S. application Ser. No. 17/377,620, filed on Jul. 16, 2021 (now U.S. Pat. No. 11,854,956), the entirety of which is incorporated by reference herein.

BACKGROUND

Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon. Many integrated circuits are typically manufactured on a single semiconductor wafer, and individual dies on the wafer are singulated by sawing between the integrated circuits along a scribe line. The individual dies are typically packaged separately, in multi-chip modules (MCM), for example, or in other types of packaging. A package (structure) not only provides protection for semiconductor devices from environmental contaminants, but also provides a connection interface for the semiconductor devices packaged therein.

One smaller type of packaging for semiconductors is a flip chip chip-scale package (FcCSP), in which a semiconductor die is placed upside-down on a substrate and bonded to the substrate using bumps. The substrate has wiring routed to connect the bumps on the die to contact pads on the substrate that have a larger footprint. An array of solder balls is formed on the opposite side of the substrate and is used to electrically connect the packaged die to an end application.

Although existing package structures and methods for fabricating package structures have generally been adequate for their intended purposes, they have not been entirely satisfactory in all respects.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1C are cross-sectional views of various stages of a process for forming a semiconductor die package in accordance with some embodiments.

FIG. 2 is a plan view of a semiconductor die package in accordance with some embodiments, wherein the passivation layer, underfill element, and the molding layer over the package substrate are omitted.

FIG. 3 is a plan view of a conductive line crack prevention design in accordance with some embodiments.

FIG. 3A is a plan view of a conductive line crack prevention design in accordance with some embodiments.

FIG. 4 is a plan view of a conductive line crack prevention design in accordance with some embodiments.

FIG. 5 is a plan view of a conductive line crack prevention design in accordance with some embodiments.

2

FIG. 6 is a plan view of a conductive line crack prevention design in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. Where applicable, the term “substantially” may also relate to 90% or higher, such as 95% or higher, especially 99% or higher, including 100%. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” are to be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word “substantially” does not exclude “completely” e.g., a composition which is “substantially free” from Y may be completely free from Y.

Terms such as “about” in conjunction with a specific distance or size are to be interpreted so as not to exclude insignificant deviation from the specified distance or size and may include for example deviations of up to 10%. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10%.

A semiconductor die package (structure) having a conductive line crack prevention design and the method of forming the same are provided in accordance with various exemplary embodiments of the disclosure. The variations of the embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. In accordance with some embodiments, a semiconductor die package utilizes non-linear or curved conductive lines at the predetermined die corner areas within the die shadow to disperse stress, thereby reducing the risk of line crack after thermal processes. Accordingly, the reliability of the semiconductor die package is improved.

Embodiments will be described with respect to a specific context, namely a chip scale package (CSP), particularly flip chip CSP (FcCSP). Other embodiments may also be applied, however, to other packaging techniques, such as flip chip ball grid array (FcbGA) packages and other packaging techniques, such as with an interposer or other active chip in a two and a half dimensional integrated circuit (2.5DIC) structure or a three dimensional IC (3DIC) structure. Although method embodiments may be discussed below as being performed in a particular order, other method embodiments contemplate steps that are performed in any logical order.

FIGS. 1A-1C illustrate cross-sectional views of intermediate stages in the formation of a semiconductor die package in accordance with some embodiments. As shown in FIG. 1A, a carrier substrate **100** is provided, in accordance with some embodiments. The carrier substrate **100** may be configured to provide temporary mechanical and structural support for the processing of build-up layers or structures during subsequent processing steps. In some embodiments, the carrier substrate **100** is a glass carrier substrate, a ceramic carrier substrate, or the like.

As shown in FIG. 1A, a release layer **102** is formed over the carrier substrate **100**, in accordance with some embodiments. The release layer **102** may be formed of a polymer-based material, and may be removed along with the carrier substrate **100** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer **102** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating. In some other embodiments, the release layer **102** is an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. The release layer **102** may be dispensed as a liquid and then cured, may be a laminate film laminated onto the carrier substrate **100**, or may be the like.

As shown in FIG. 1A, a package substrate **104** is formed on the release layer **102**, in accordance with some embodiments. The package substrate **104** has a surface **104A** facing the release layer **102**, and another surface **104B** opposite to the surface **104A**. The package substrate **104** may be used for routing and configured to provide electrical connection between semiconductor devices packaged in the package structure and an external electronic device such as a printed circuit board (PCB), which is described in further detail below.

In FIG. 1A, the package substrate **104** is a core-less redistribution substrate (but the disclosure is not limited thereto), which includes conductive pads **1041** formed over the release layer **102**, a dielectric layer **1042** formed over the release layer **102** and the conductive pads **1041**, a wiring layer **1043** formed over the dielectric layer **1042** and in through holes **1042a** of the dielectric layer **1042**, a dielectric layer **1044** formed over the dielectric layer **1042** and the wiring layer **1043**, a wiring layer **1045** formed over the dielectric layer **1044** and in through holes **1044a** of the dielectric layer **1044**, a dielectric layer **1046** formed over the dielectric layer **1044** and the wiring layer **1045**, a wiring layer **1047** formed over the dielectric layer **1046** and in through holes **1046a** of the dielectric layer **1046**, a dielectric layer **1048** formed over the dielectric layer **1046** and the wiring layer **1047**, and conductive pads **1049** formed over the dielectric layer **1048** and in through holes **1048a** of the dielectric layer **1048**, in accordance with some embodiments.

Each of the wiring layers **1043**, **1045**, and **1047** includes conductive lines providing electrical connection in horizon-

tal directions and conductive vias providing electrical connection in vertical directions, and the wiring layers **1043**, **1045**, and **1047** are electrically connected to each other, in accordance with some embodiments. The conductive pads **1041** and **1049** may be exposed at or protruding from the surfaces **104A** and **104B** of the package substrate **104**, respectively, and are electrically connected to the wiring layers **1043**, **1045**, and **1047**, in accordance with some embodiments.

The dielectric layers **1042**, **1044**, **1046**, and **1048** may be made of or include an insulating material such as a polymer material (e.g., polybenzoxazole, polyimide, or a photosensitive material), nitride (e.g., silicon nitride), oxide (e.g., silicon oxide), silicon oxynitride, or the like. The wiring layers **1043**, **1045**, and **1047** and the conductive pads **1041** and **1049** may be made of or include copper, aluminum, gold, cobalt, titanium, nickel, silver, graphene, one or more other suitable conductive materials, or a combination thereof. The formation of the package substrate **104** (i.e., redistribution substrate) may involve multiple deposition or coating processes, multiple patterning processes, and/or multiple planarization processes. These processes are well known in the art and therefore not described herein.

It should be appreciated that the package substrate **104** shown in FIG. 1A is merely an illustrative example, and many variations and/or modifications can be made to embodiments of the disclosure. For example, the structure, configuration, location, and/or number of the dielectric layers, wiring layers, and/or conductive pads of the package substrate **104** can be changed in different embodiments.

In some other embodiments, the package substrate **104** further includes a core layer (not shown). The core layer may be made of or include a fiber material, a polymer material, a semiconductor material, a glass material, a metal material, or another suitable material. Interconnect structures (for example, composed of dielectric layers, wiring layers, and conductive pads, similar to those previously illustrated in FIG. 1A) can be formed on both sides of the core layer to facilitate routing.

In some embodiments, the package substrate **104** is an interposer substrate which is free from active devices (e.g., transistors and diodes) and passive devices (e.g., resistors, capacitors, inductors, or the like) therein. In some alternative embodiments, the package substrate **104** is a device substrate which includes active and/or passive devices therein.

In accordance with some embodiments, a passivation layer **105** is further formed on the surface **104B** of the package substrate **104**, as shown in FIG. 1A. The passivation layer **105** (sometimes also called a solder mask) may be configured to protect the underlying metal lines from damage of moisture or other detrimental chemicals. In some embodiments, the passivation layer **105** is formed of or includes inorganic materials such as silicon nitride, silicon oxide, or multi-layers of the organic or inorganic materials. A patterning process is then performed to forming openings **105a** (see FIG. 1A) in the passivation layer **105** to partially expose the underlying conductive pads **1049**, in accordance with some embodiments. The patterning process may include a photolithography process, an energy beam drilling process (such as a laser beam drilling process, an ion beam drilling process, or an electron beam drilling process), an etching process, a mechanical drilling process, one or more other applicable processes, or a combination thereof. In some other embodiments, the passivation layer **105** can be omitted.

As shown in FIG. 1B, one or more semiconductor dies **106** (for illustration, only one semiconductor die **106** is

shown) are stacked over the surface **104B** of the package substrate **104** using, for example, a pick-and-place tool (not shown), in accordance with some embodiments. The semiconductor die **106** may be a logic die (e.g., central processing unit, graphics processing unit, field-programmable gate array (FPGA), system-on-chip (SOC) die, microcontroller, or the like), a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, or the like), a power management die (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) die), or the like. Before being disposed over the package substrate **104**, the semiconductor die **106** may be processed according to applicable manufacturing processes to form integrated circuits in the semiconductor die **106**. The processes are well known in the art and therefore not described herein.

In some embodiments, the semiconductor die **106** includes a semiconductor substrate (not individually shown), such as silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate may also include other semiconductor material, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used.

In some embodiments, devices (not shown) forming integrated circuits of the semiconductor die **106**, such as transistors, diodes, capacitors, resistors, etc., may be formed on or in the semiconductor substrate. A passivation layer (not shown) may be formed over semiconductor substrate to cover the devices.

As shown in FIG. 1B, the semiconductor die **106** also includes conductive pads **1061** exposed at the active surface **106A** (e.g., the bottom surface as shown) for external connections, in accordance with some embodiments. The conductive pads **1061** may be embedded in the passivation layer and are electrically connected to the integrated circuits in the semiconductor die **106**. In some embodiments, the conductive pads **1061** are made of or include a conductive material, such as metal (e.g., copper, aluminum, nickel, or combinations thereof).

As shown in FIG. 1B, the semiconductor die **106** is bonded to the conductive pads **1049** of the package substrate **104**, exposed through the passivation layer **105**, for example, through flip-chip bonding by using conductive connectors **107**, in accordance with some embodiments. The conductive connectors **107** may include solder balls, micro bumps, or the like, and may be made of or include a solder material. In some embodiments, the solder material may be formed on the exposed conductive pads **1061** of the semiconductor die **106**, formed on the exposed conductive pads **1049** of the package substrate **104**, or formed on both before bonding, such as using an electroplating process, an electroless plating process, a placement process, a printing process, a physical vapor deposition (PVD) process, a chemical vapor deposition (CVD) process, a photolithography process, one or more other applicable processes, or a combination thereof. A reflow process is then performed to melt the solder material to form the conductive connectors

107, to physically and electrically couple the semiconductor die **106** to the package substrate **104**, in accordance with some embodiments.

As shown in FIG. 1B, an underfill element **108** is formed over the surface **104B** of package substrate **104** (such as in direct contact with the passivation layer **105**) to surround and protect the semiconductor die **106** and the conductive connectors **107** thereunder, and enhances the connection between the semiconductor die **106** and the package substrate **104**, in accordance with some embodiments. The underfill element **108** may be made of or include an insulating material such as an underfill material. The underfill material may include an epoxy, a resin, a filler material, a stress release agent (SRA), an adhesion promoter, another suitable material, or a combination thereof. In some embodiments, an underfill material in liquid state is dispensed into the gap between the semiconductor die **106** and the passivation layer **105** to reinforce the strength of the conductive connectors **107** and therefore the overall package structure. After the dispensing, the underfill material is cured (for example by heating or ultraviolet (UV) radiation) to form the underfill element **108**, in accordance with some embodiments.

As shown in FIG. 1B, a molding layer **109** is formed over the surface **104B** of package substrate **104** (such as in direct contact with the passivation layer **105**) to encapsulate and protect the semiconductor die **106** and the underfill element **108**, in accordance with some embodiments. The molding layer **109** may be separated from the conductive connectors **107** below the semiconductor die **106** by the underfill element **108**. In some embodiments, the molding layer **109** is made of or includes an insulating material such as a molding material. The molding material may include a polymer material, such as an epoxy-based resin with fillers dispersed therein. In some embodiments, a molding material (such as a liquid molding material) is dispensed using, for example, injection molding over the passivation layer **105**, the semiconductor die **106**, and the underfill element **108**. In some other embodiments, the underfill element **108** is omitted, and the molding material may be dispensed into the gap between the semiconductor die **106** and the passivation layer **105** (or the package substrate **104**) and in direct contact with the conductive connectors **107** below the semiconductor die **106**. A thermal process is then used to cure the liquid molding material and to transform it into the molding layer **109**, in accordance with some embodiments.

Because there is a coefficient of thermal expansion (CTE) mismatch between the package substrate **104** and the devices or other materials thereon, the warpage of the package substrate **104** occurs during the thermal processes as described above. The bending can cause reliability issues, such as crack in conductive lines of the package substrate **104** due to stress. It has been observed that the stress is typically concentrated in the areas between (or corresponding) corners of the semiconductor die **106** and package substrate **104**, which causes cracks to easily occur in the conductive lines of the package substrate **104** in these areas. In accordance with some embodiments, as shown in FIG. 2, the stress concentration areas SA (slash areas shown) in the package substrate **104** are close to the corners C of the semiconductor die **106** (therefore, hereinafter may also referred to as the die corner areas) and are under the shadow of the semiconductor die **106** (i.e., within the projection area PA (see FIG. 1B) of the semiconductor die **106**). In some embodiments, each stress concentration area SA is square with the length L_1 , L_2 of each side of about $500\text{ }\mu\text{m}$ (i.e., with an area of about $250,000\text{ }\mu\text{m}^2$), but the disclosure is not

limited thereto. For example, the shape and size of the stress concentration areas SA can vary in different embodiments, depending on the configuration, shape, size, material, process, etc. for the various components in the package structure.

Various embodiments related to conductive line crack prevention design for preventing cracks in conductive lines (particularly the line parts within the stress concentration areas SA) of the package substrate **104** are described in the following. FIGS. 3-5 illustrate plan views of conductive line crack prevention designs in accordance with some embodiments of the disclosure. It should be appreciated that those conductive line crack prevention designs may be applied to one or more wiring layers in the package structure (such as the wiring layers **1043**, **1045**, and **1047** of the package substrate **104** shown in FIG. 1B) to prevent cracks in the conductive line(s) thereof.

As shown in FIG. 3, the conductive line with crack prevention design includes connected line segments **301** and **302**, in accordance with some embodiments. In some embodiments, the line segments **301** and **302** (and their intersection S_1) are located in each stress concentration area (die corner area) SA under the shadow of the semiconductor die **106** (see FIG. 2). In some other embodiments, the entire line segment **302** is located in each stress concentration area SA, and a portion of the line segment **301** is outside the stress concentration area SA.

In some embodiments, the line segment **301** is a straight line (i.e., linear) and extends in a first direction D_1 . The first direction D_1 may be a horizontal direction substantially parallel to the surface **104B** of the package substrate **104** (see FIG. 1B), and may be parallel to, perpendicular to an outer side **104C** (see FIG. 2) of the package substrate **104** or form an angle with the outer side **104C**. In some embodiments, the line segment **301** is configured to electrically connect to one of the conductive pads **1049** or **1041** (see FIG. 1B) or electrically connect to an overlying or underlying wiring layer (such as the wiring layer **1043**, **1045**, or **1047** shown in FIG. 1B) through a conductive via therebetween. For example, as shown in FIG. 3, one end **301A** of the line segment **301** may be connected to one conductive pad **1049/1041** or one conductive via CV_1 between two adjacent wiring layers of the package structure, and the other end **301B** may be connected to the line segment **302**.

In some embodiments, the line segment **302** is configured for routing in the package substrate **104**. In some embodiments, the line width W_2 of the line segment **302** is smaller than the line width W_1 of the line segment **301**. The ratio of the line width W_2 to the line width W_1 may be about 1:1.5 to 1:2, but other ratios can also be used. The line segment **302** with a smaller width can save space for arranging more conductive lines (i.e., higher wiring density). It should be appreciated that the stress tends to concentrate at the intersection S_1 of the line segment **301** and the line segment **302** with different line widths, so cracks usually occur in the intersection S_1 . To this end, in accordance with some embodiments, the line segment **302** is routed in a non-linear pattern with a varying extension direction (i.e., different parts of the line segment **302** extend in various directions) to disperse stress and avoid the stress from being concentrated at the intersection S_1 , thereby reducing the risk of line cracks.

In some embodiments, one end **302A** of the line segment **302** is connected to the end **301B** of the line segment **301** to form the intersection S_1 , and the other end **302B** is away from the line segment **301**. In some embodiments, the line segment **302** has a zig-zag (serpentine) pattern and consists

of several S-shaped lines. For example, in the embodiments of FIG. 3, the line segment **302** consists of four connected S-shaped lines **3021**, **3022**, **3023**, and **3024**. The S-shaped line **3024** is closest to the intersection S_1 , the S-shaped line **3021** is farthest from the intersection S_1 , and the S-shaped lines **3022** and **3023** are between the S-shaped lines **3021** and **3024**. One of ordinary skill in the art will appreciate that other numbers (for example, one, two, three or more than four, etc.) of the S-shaped lines can also be used.

In some embodiments, as shown in FIG. 3, both ends **302A** and **302B** of the line segment **302** are located on a virtual straight line VL (it can also be regarded as a virtual connecting line of the ends **302A** and **302B**). The virtual straight line VL is aligned with the line segment **301** in the first direction D_1 . In this case, each S-shaped line of the line segment **302** can be composed of two connecting U-shaped curves (bending parts) that are on both sides of the virtual straight line VL. The two U-shaped curves of each S-shaped line may have the same shape/size or different shapes/sizes. For example, in FIG. 3, the two U-shaped curves of the S-shaped lines **3022** and **3023** are symmetrical U-shaped and have the same size, and the two U-shaped curves of the S-shaped lines **3021** and **3024** include a symmetrical U-shaped curve and an asymmetrical U-shaped curve, with different sizes (for example, the width (end-to-end distance) of the asymmetrical U-shaped curve (of S-shaped line **3021** or **3024**) adjacent to the intersection S_1 or S_2 in the first direction D_1 is greater than the width of the symmetrical U-shaped curve (of S-shaped line **3021** or **3024**) adjacent to the S-shaped line **3022** or **3023** in the first direction D_1). That is to say, the shape of the S-shaped line **3021** is different from the S-shaped lines **3022**, **3023** and **3024**, and the shape of the S-shaped line **3024** is different from the S-shaped lines **3021**, **3022** and **3023**. In some other embodiments, the virtual straight line VL may also form an angle greater than 0 degrees with the first direction D_1 .

As shown in FIG. 3, an angle α is formed between a tangent direction T of a part of the line segment **302** (i.e., the S-shaped line **3024**) that intersects the line segment **301** and a second direction D_2 perpendicular to the first direction D_1 . In accordance with some embodiments, the formed angle α is greater than 0 degrees and equal to or less than 60 degrees (i.e., $0^\circ < \alpha \leq 60^\circ$), so that the line segment **302** can effectively disperse stress and avoid stress concentration at the intersection S_1 . If the angle α is equal to 0 degrees, a right angle is formed between the line segment **301** and the line segment **302**, where the stress can be easily concentrated to cause cracks. If the angle α is greater than 60 degrees, the pattern of the line segment **302** is close to a straight line, thus losing the ability to disperse stress.

In some embodiments, as shown in FIG. 3, the maximum size P (i.e., the distance between the leftmost and rightmost ends shown in the figure) of the line segment **302** in the second direction D_2 is greater than the line width W_1 of the line segment **301** in the second direction D_2 . In some embodiments, in order to reduce the occupied space, the maximum size P of the line segment **302** does not exceed twice the line width W_1 of the line segment **301**. In addition, the distance X (see FIG. 3) between the ends **302A** and **302B** of the line segment **302** is equal to or less than about 100 μm to save space, in accordance with some embodiments. The length of the line segment **302** in the first direction D_1 (i.e., the distance X) may generally be greater than the length L of the line segment **301** in the first direction D_1 , but the disclosure is not limited thereto.

As shown in FIG. 3, the line segment **302** is further connected to a line segment **303** and is interposed between

the line segment **301** and the line segment **303**, in accordance with some embodiments. The line segment **303** may be configured for routing in the package substrate **104**, similar to the line segment **302**. In some embodiments, the line width W_3 of the line segment **303** is substantially equal to the line width W_2 of the line segment **302** and smaller than the line width W_1 of the line segment **301**, so as to save space. In some embodiments shown in FIG. 3, the line segment **303** is a straight line (i.e., linear) and extends in the first direction D_1 . In some other embodiments, the linear line segment **303** extends in a direction different from the first direction D_1 (i.e., an angle greater than 0 degrees is formed between the linear line segment **303** and the first direction D_1). In some alternative embodiments shown in FIG. 3A, the line segment **303** is non-linear and has at least one bend.

In some embodiments, one end **303A** of the line segment **303** is connected to the end **302B** of the line segment **302** to form an intersection **S 2** (which may be located in the stress concentration area **SA**), and the other end **303B** is away from the line segment **302**. The end **303B** of the line segment **303** may be configured to electrically connect to one of the conductive pads **1049** or **1041** (see also FIG. 1B) or electrically connect to an overlying or underlying wiring layer (such as the wiring layer **1043**, **1045**, or **1047** shown in FIG. 1B) through a conductive via **CV 2** therebetween.

Many variations and/or modifications can be made to embodiments of the disclosure. For example, the shape or pattern of the narrower line segment **302** connecting the wider line segment **301** can vary in different embodiments.

FIG. 4 illustrates a plan view of a conductive line crack prevention design in accordance with some alternative embodiments, wherein the line segment **402** (interposed between the line segments **301** and **303**) in FIG. 4 has a mirror pattern of the line segment **302** of the embodiments in FIG. 3. Other structures in FIG. 4 are the same as those in FIG. 3.

FIG. 5 illustrates a plan view of a conductive line crack prevention design in accordance with some other embodiments. The most of the structures in FIG. 5 are the same as those in FIG. 3, so only the different parts are described here. In FIG. 5, the narrower line segment **502** connecting the wider line segment **301** has a zig-zag pattern and consists of several lightning-shaped lines (such as three lightning-shaped lines **5021**, **5022**, and **5023**, but other numbers can also be used). Each of the lightning-shaped lines **5021**, **5022**, and **5023** is composed of multiple straight lines extending in different directions than the first direction D_1 . An angle β of about 30 degrees to about 60 degrees may be formed at the intersection of two connected lines in some cases, but other degrees can also be used. Alternatively, each of the lightning-shaped lines **5021**, **5022**, and **5023** can be composed of two triangular bending parts that are on both sides of the virtual straight line **VL**, as shown in FIG. 5. With the above design, it can also help disperse stress and avoid stress concentration at the intersection of line segments with different line widths, thereby reducing the risk of line cracks.

FIG. 6 illustrates a plan view of a conductive line crack prevention design in accordance with some other embodiments. The most of the structures in FIG. 6 are the same as those in FIG. 3, so only the different parts are described here. In FIG. 6, the narrower line segment **602** connecting the wider line segment **301** is routed in a curved pattern. In accordance with some embodiments, the curved pattern is a parabolic pattern. In this case, the line segment **602** has one bending part on one side of the virtual straight line **VL**, as shown in FIG. 6. However, other applicable curved patterns can also be used. For example, the curved pattern may

include multiple bending parts on one side or both sides of the virtual straight line **VL**, in some other embodiments. As shown in FIG. 6, an angle α is formed between a tangent direction **T** of a part of the line segment **602** that intersects the line segment **301** and the second direction D_2 . In accordance with some embodiments, the formed angle α is greater than 0 degrees and equal to or less than 60 degrees (i.e., $0^\circ < \alpha \leq 60^\circ$), so that the line segment **602** can effectively disperse stress and avoid stress concentration at the intersection **S₁** (similar to the line segment **302** illustrated in FIG. 3).

One of ordinary skill in the art will appreciate that the above conductive line crack prevention design examples are provided for illustrative purposes, and other designs utilizing different routing patterns (as long as they can be formed using the processes described above or any other known techniques) that can avoid stress concentration at the intersection of line segments with different line widths may also be used.

Referring back to FIG. 1B, after forming the molding layer **109**, a planarization process is further performed to partially remove the molding layer **109**, in accordance with some embodiments. As a result, a thickness of the molding layer **109** in a vertical direction perpendicular to the surface **104B** is reduced, and the overall package structure is thinner.

In some embodiments, the top surface **106B** of the semiconductor die **106** is exposed from the molding layer **109** (for example, the top surface **106B** is substantially flush with the top surface **109A** of the molding layer **109**) after the planarization process, as shown in FIG. 1B. This facilitates the dissipation of heat generated from the semiconductor die **106** during operation. In some other embodiments, the semiconductor die **106** is still buried in (i.e., the top surface **106B** being covered by a portion of the molding layer **109**) the molding layer **109** after the planarization process. The planarization process may include a grinding process, a chemical mechanical polishing (CMP) process, an etching process, a dry polishing process, one or more other applicable processes, or a combination thereof.

As shown in FIG. 1C, the carrier substrate **100** and the release layer **102** are removed to expose the surface **104A** of the package substrate **104**, in accordance with some embodiments. Afterwards, conductive bumps **110** are formed over the surface **104A** that is originally covered by the carrier substrate **100**, in accordance with some embodiments. Each of the conductive bumps **110** may be electrically connected to one of the conductive pads **1041** exposed at the surface **104A**. In some embodiments, the conductive bumps **110** are or include solder bumps such as tin-containing solder bumps. The tin-containing solder bumps may further include copper, silver, gold, aluminum, lead, one or more other suitable materials, or a combination thereof. In some embodiments, the tin-containing solder bump is lead-free.

In some embodiments, solder balls (or solder elements) are disposed on the exposed conductive pads **1041** after the removal of the carrier substrate **100**. A reflow process is then carried out to melt the solder balls into the conductive bumps **110**. In some other embodiments, under bump metallization (UBM) elements are formed over the exposed conductive pads **1041** before the solder balls are disposed. In some other embodiments, solder elements are electroplated onto the exposed conductive pads **1041**. Afterwards, a reflow process is used to melt the solder element to form the conductive bumps **110**. The conductive bumps **110** (sometimes also called ball grid array (BGA)) allow the semiconductor die package structure to be bonded and electrically connected to

an external PCB or other electronic devices (not shown), in accordance with some embodiments.

As a result, the process for forming the resulting semiconductor die package structure illustrated in FIG. 1C is completed. It should be appreciated that the above method examples are provided for illustrative purposes, and other methods that can form the resulting package structure of FIG. 1C can also be used.

Although the above package structure includes only one semiconductor die, two or more semiconductor dies of the same or different types can also be packaged in the package structure, and the above conductive line crack prevention designs can be used to prevent the conductive lines of the package substrate under the semiconductor dies at the predetermined stress concentration areas (die corner areas) from cracking after thermal processes. Additionally or alternatively, the package structure may also include a package module including an interposer substrate and one or more semiconductor dies mounted thereon, in some other embodiments. The package module is bonded to the package substrate by, for example, flip-chip bonding. The above conductive line crack prevention designs can also be used to prevent the conductive lines of the package substrate under the package module at the predetermined stress concentration areas from cracking after thermal processes. Furthermore, in accordance with some other embodiments, the linear line segment (such as the line segment 301) and the non-linear or curved line segment (such as the line segment 302, 402, or 502) of the above conductive line crack prevention designs have the same line width.

Embodiments of the disclosure form a semiconductor die package including a package substrate and a semiconductor die disposed over the package substrate. In accordance with some embodiments, the conductive lines of at least one wiring layer of the package substrate at the predetermined die corner areas (stress concentration areas) under the die shadow are routed in a non-linear or curved pattern to disperse stress and avoid stress concentration at the intersection of line segments with different line widths, thereby reducing the risk of line crack after thermal processes. Accordingly, the reliability of the semiconductor die package is improved.

In accordance with some embodiments, a semiconductor die package is provided. The semiconductor die package includes a semiconductor die and a package substrate supporting and electrically connected to the semiconductor die. The semiconductor die has a corner. The package substrate includes several conductive lines, and one of the conductive lines under the corner of the semiconductor die includes a first line segment and a second line segment connected to the first line segment. The first line segment is linear and extends in a first direction. The second line segment is non-linear and has a varying extension direction.

In accordance with some embodiments, a semiconductor die package is provided. The semiconductor die package includes a semiconductor die and a package substrate supporting and electrically connected to the semiconductor die. The semiconductor die has a corner. The package substrate includes several conductive lines, and one of the conductive lines under the corner of the semiconductor die includes a first line segment and a second line segment connected to the first line segment. The first line segment is linear. The second line segment is non-linear and has at least one bending part.

In accordance with some embodiments, a semiconductor die package is provided. The semiconductor die package includes a semiconductor die and a package substrate dis-

posed below the semiconductor die. The semiconductor die has a corner. The package substrate includes several conductive lines, and one of the conductive lines under the corner of the semiconductor die includes a first line segment, a second line segment, and a third line segment. The second line segment is interposed between the first line segment and the third line segment. The first line segment is linear. The second line segment is non-linear and has at least one bending part.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor die package, comprising:

a semiconductor die having a corner; and
a package substrate supporting and electrically connected to the semiconductor die, wherein the package substrate comprises a plurality of conductive lines, and one of the conductive lines under the corner comprises:
a first line segment; and
a second line segment connected to the first line segment, wherein the first line segment is linear and extends in a first direction, and the second line segment is non-linear and has a varying extension direction.

2. The semiconductor die package as claimed in claim 1, wherein the second line segment is located in a die corner area which is within a projection area of the semiconductor die and close to the corner.

3. The semiconductor die package as claimed in claim 1, wherein a line width of the second line segment is smaller than a line width of the first line segment.

4. The semiconductor die package as claimed in claim 1, wherein an angle formed between a tangent direction of a part of the second line segment that intersects the first line segment and a second direction perpendicular to the first direction is greater than 0 degrees and equal to or less than 60 degrees.

5. The semiconductor die package as claimed in claim 1, wherein a distance between a leftmost end of the second line segment and a rightmost end of the second line segment in a second direction is greater than a line width of the first segment in the second direction, the second direction being perpendicular to the first direction.

6. The semiconductor die package as claimed in claim 1, wherein the second line segment has a parabolic pattern.

7. The semiconductor die package as claimed in claim 1, wherein the second line segment has a zig-zag pattern.

8. The semiconductor die package as claimed in claim 1, wherein the second line segment has a first end connected to the first line segment and a second end away from the first line segment, and the first end and the second end are located on a virtual straight line aligned with the first line segment in the first direction.

9. Semiconductor die package as claimed in claim 1, wherein the package substrate further comprises a plurality of conductive pads through which the semiconductor die is

13

bonded to the package substrate, wherein the first line segment has a first end connected to one of the conductive pads and a second end connected to the second line segment.

10. The semiconductor die package as claimed in claim 1, wherein the package substrate further includes a plurality of conductive vias electrically connected to some of the conductive lines stacked in a vertical direction perpendicular to the first direction, wherein the first line segment has a first end connected to one of the conductive vias and a second end connected to the second line segment.

11. A semiconductor die package, comprising:
a semiconductor die having a corner; and

a package substrate supporting and electrically connected to the semiconductor die, wherein the package substrate comprises a plurality of conductive lines, and one of the conductive lines under the corner comprises:
a first line segment; and
a second line segment connected to the first line segment, wherein the first line segment is linear, and the second line segment is non-linear and has at least one bending part.

12. The semiconductor die package as claimed in claim 11, wherein the second line segment is located in a die corner area which is within a projection area of the semiconductor die and close to the corner.

13. The semiconductor die package as claimed in claim 11, wherein the second line segment has a first end connected to the first line segment and a second end away from the first line segment, and the first end and the second end are located on a virtual straight line, wherein the at least one bending part of the second line segment is located on at least one side of the virtual straight line.

14. The semiconductor die package as claimed in claim 13, wherein the second line segment has a parabolic pattern and has one bending part on one side of the virtual straight line.

14

15. The semiconductor die package as claimed in claim 13, wherein the second line segment has a zig-zag pattern and has bending parts on both sides of the virtual straight line.

16. A semiconductor die package, comprising:
a semiconductor die having a corner; and

a package substrate below the semiconductor die and comprising a plurality of conductive lines, wherein one of the conductive lines under the corner comprises:

a first line segment;

a second line segment; and

a third line segment, wherein the second line segment is interposed between the first line segment and the third line segment, and is located in a die corner area which is within a projection area of the semiconductor die and close to the corner, and wherein the first line segment is linear, and the second line segment is non-linear and has at least one bending part, wherein the first line segment extends in a first direction, and wherein an angle formed between a tangent direction of a part of the second line segment that intersects the first line segment and a second direction perpendicular to the first direction is greater than 0 degrees and equal to or less than 60 degrees.

17. The semiconductor die package as claimed in claim 16, wherein the third line segment is linear and extends in the first direction.

18. Semiconductor die package as claimed in claim 16, wherein the third line segment is non-linear and has at least one bending part.

19. The semiconductor die package as claimed in claim 16, wherein the second line segment has a zig-zag pattern.

20. The semiconductor die package as claimed in claim 16, wherein a line width of the second line segment is smaller than a line width of the first line segment.

* * * * *